CUSTOMER CHURN ANALYSIS

Data Science Internship Project

INTRODUCTION & OBJECTIVE

- Customer churn refers to customers who stop using a service.
- Retaining customers is more cost-effective than acquiring new ones.

Objective:

- Analyze customer churn behavior
- Identify key factors influencing churn
- Suggest strategies to improve customer retention

DATASET & DATA CLEANING

- 7043 customers, 21 features
  Includes demographics, services, billing.
- Target variable: Churn

Cleaning:

- Converted TotalCharges
- Removed missing values
- Converted Churn to binary

OVERALL CHURN RATE

- 26.6% customers churned
- 73.4% retained

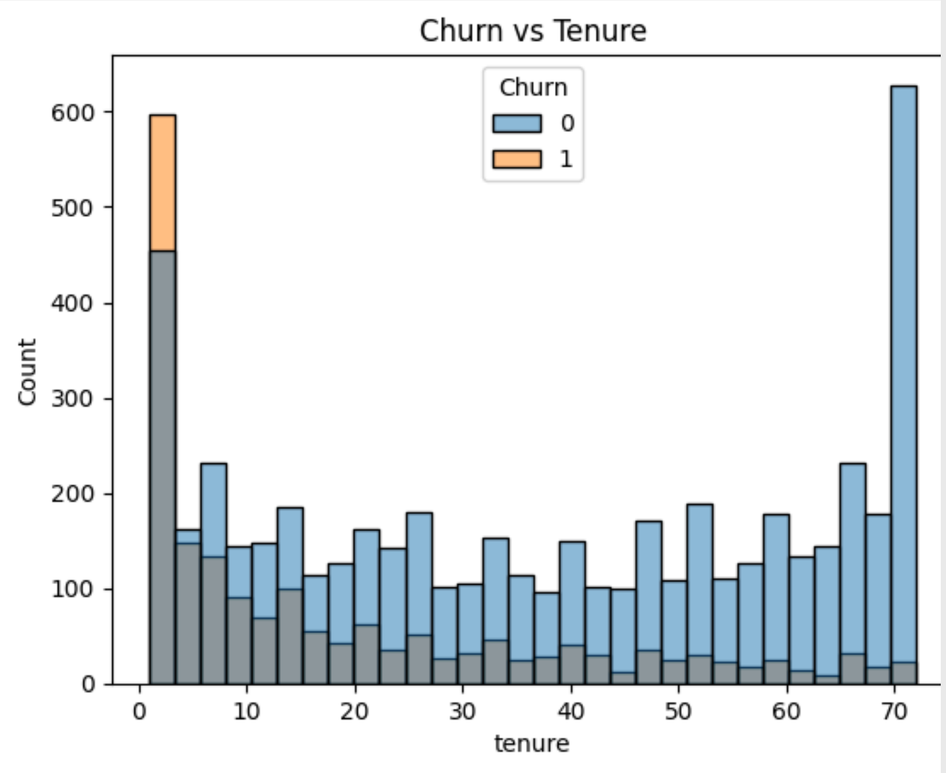
Nearly 1 in 4 customers leave

- This indicates a moderate churn rate requiring attention.

TENURE VS CHURN

- CUSTOMERS WITH LOWER TENURE CHURN MORE
- NEW USERS ARE MORE LIKELY TO LEAVE EARLY

→ EARLY CUSTOMER EXPERIENCE IS CRITICAL FOR RETENTION

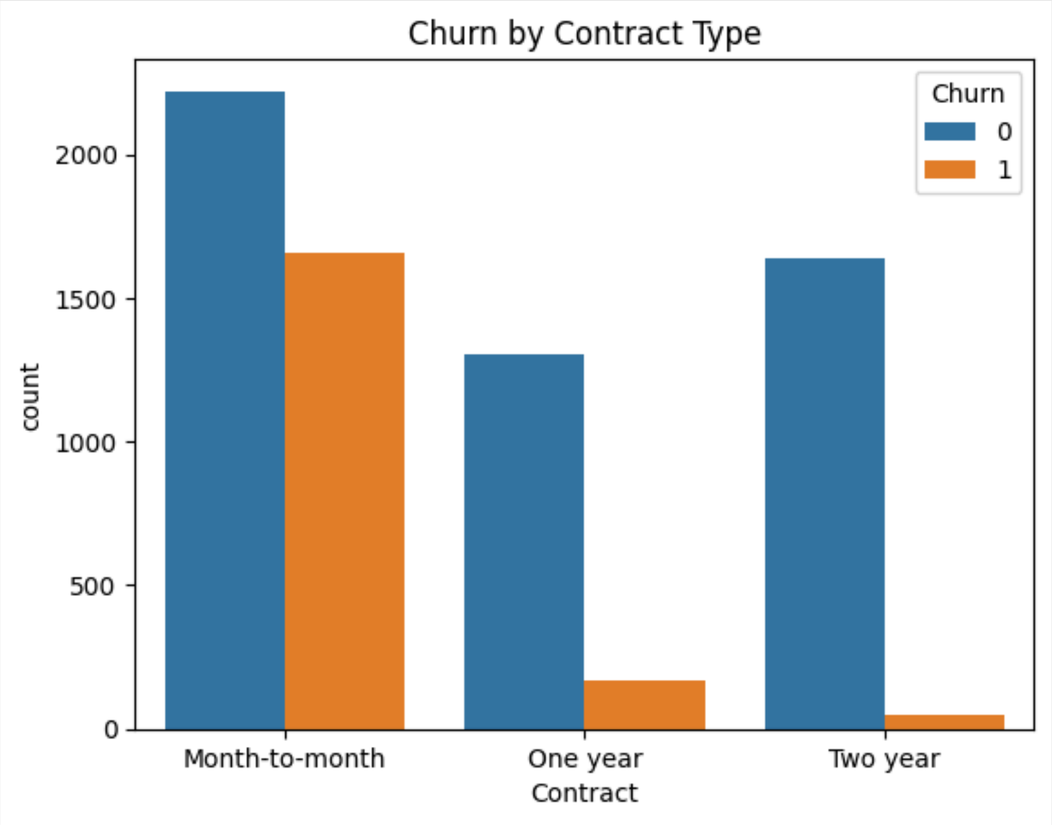


THIS INDICATES THAT CUSTOMERS ARE MOST LIKELY TO CHURN DURING THE INITIAL STAGES OF THEIR SUBSCRIPTION.

CONTRACT VS CHURN

- Month-to-month customers churn more
- Long-term contracts reduce churn

Customer commitment strongly impacts retention .

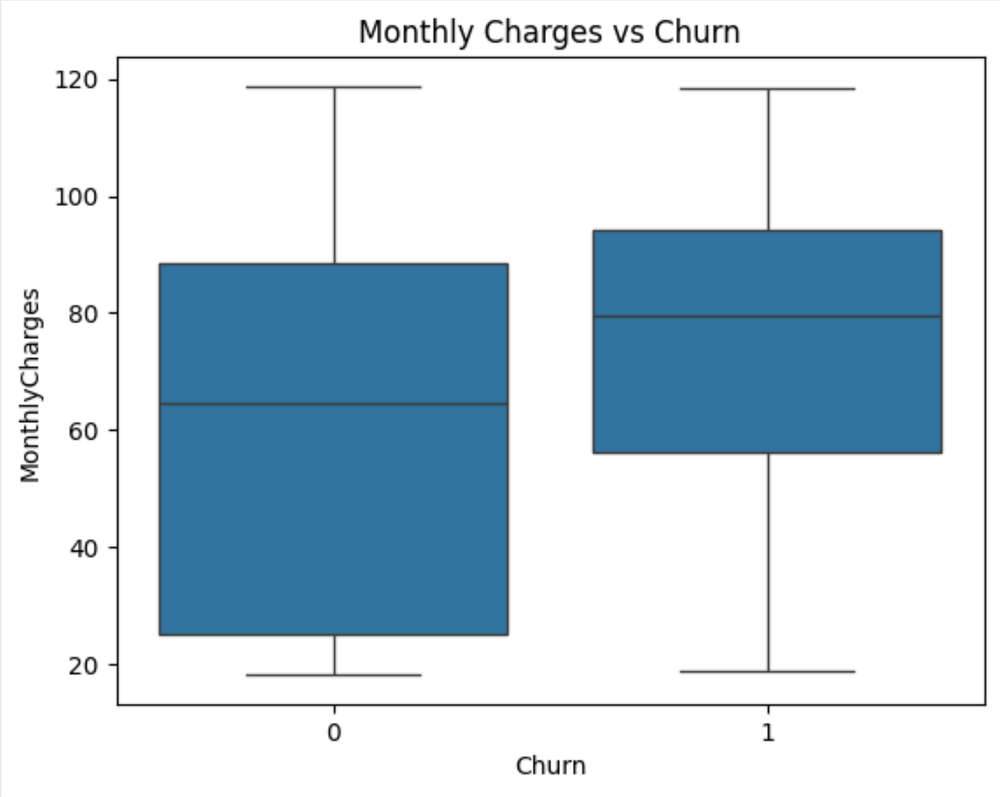


Month-to-month customers show significantly higher churn compared to long-term contracts.

MONTHLY CHARGES VS CHURN

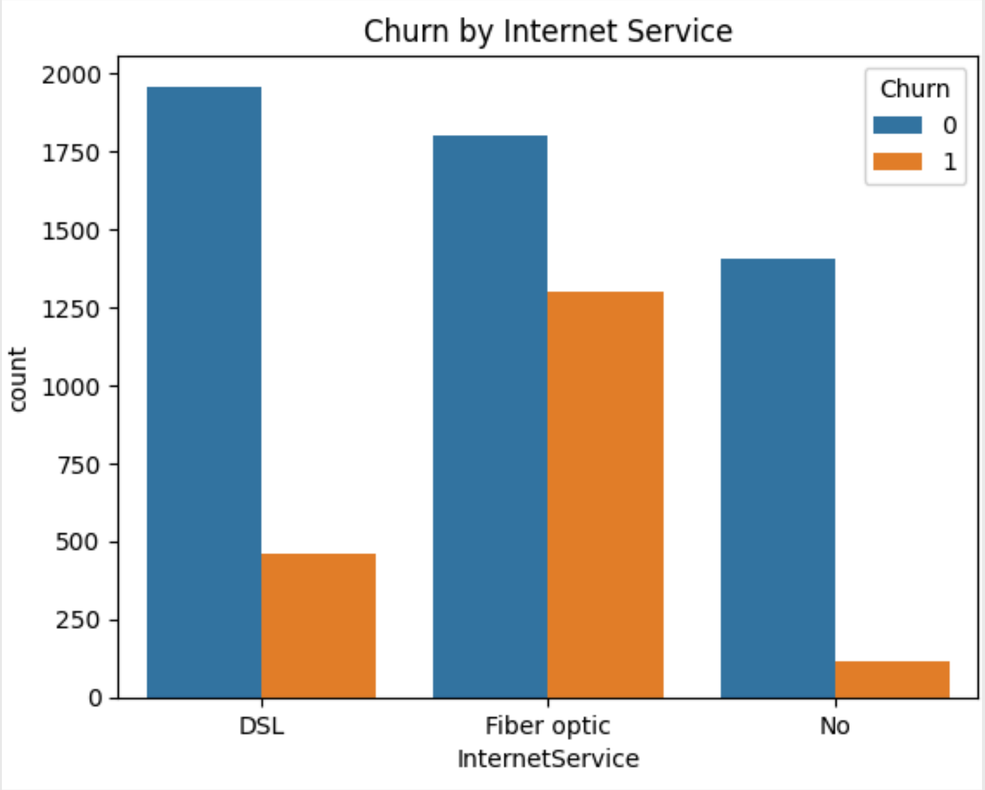
- Higher charges → higher churn
- Churned users pay more on average.

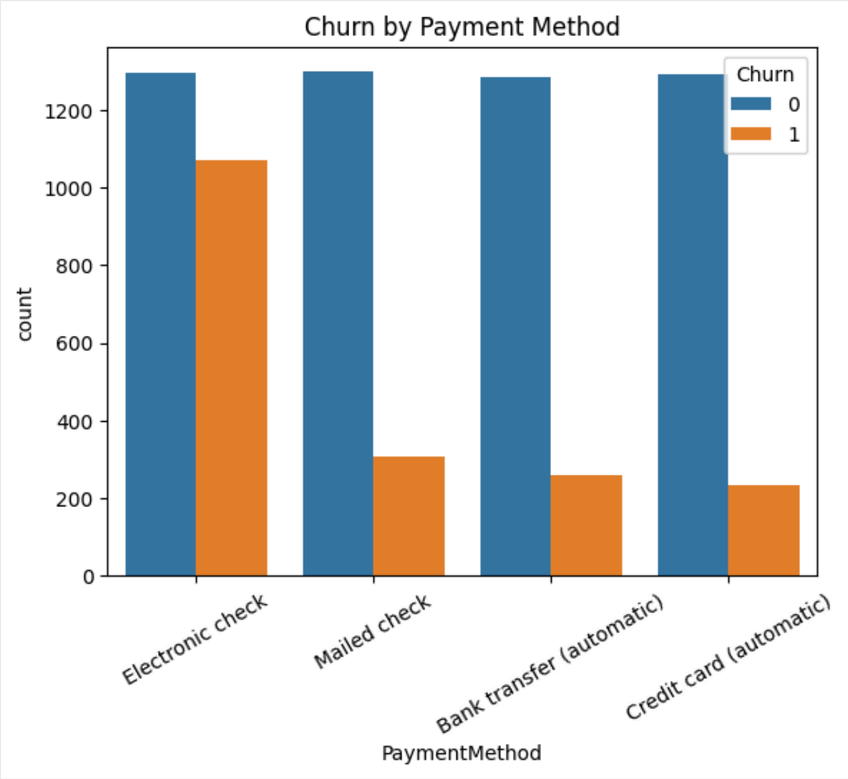
Pricing significantly influences customer decisions.



Customers with higher monthly charges are more likely to churn due to pricing sensitivity.

SERVICE & PAYMENT INSIGHTS





Fiber optic users and electronic check users show higher churn, indicating service experience and payment method impact retention.

INSIGHTS & RECOMMENDATIONS

- High-risk: Month-to-month users.
- High charges increase churn.

Recommendations:

- Offer long-term discounts
- Improve pricing
- Better onboarding

These strategies can significantly reduce customer churn.

THANK YOU

Presented by
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